

Lab Exercise

- **Make a *chatting room* using `poll()`**
 - The clients should insert the user's **username**
 - The client should be under an infinite loop with **conditional exit** ("/quit")
 - When a client sends a message, **the server receives the message and sends it to the other clients**
 - When a client joins or leaves the chatting room, server sends a message to the other clients that client has joined/left.
 - Set events for listenfd and all connfds to POLLIN.
 - Check if the POLLIN bit of revents is set for a specific fd to check if there is input (i.e., fd_struct.revents & POLLIN)
 - **Hint: poll() is needed on both the client and the server.**
 - **Timeout is not needed**
 - » e.g., `poll(fds, nfds, -1);`

Lab Exercise

■ Running example

```
spl on ● SPL in 2025s/w12/week12 via C v13.3.0-gcc
■ ./w12_client localhost 1398
Enter your username: Cthon98
AzureDiamond has joined the chat.
hey, if you type in your pw, it will show as stars
***** see!
<AzureDiamond> hunter2
<AzureDiamond> doesnt look like stars to me
<AzureDiamond> *****
thats what I see
... has joined the chat.
<AzureDiamond> oh, really?
Absolutely
See, when YOU type hunter2, it shows to us as *****
<AzureDiamond> thats neat, I didnt know chatting applications did that
yep, no matter how many times you type hunter2, it will show to us as *****
<AzureDiamond> awesome!
<AzureDiamond> wait, how do you know my pw?
... has left the chat.
er, I just copy pasted YOUR *****'s and it appears to YOU as hunter2 cause its your pw
<AzureDiamond> oh, ok.
```

```
spl on ● SPL in 2025s/w12/week12 via C v13.3.0-gcc
■ ./w12_client localhost 1398
Enter your username: ...
<AzureDiamond> oh, really?
<Cthon98> Absolutely
<Cthon98> See, when YOU type hunter2, it shows to us as *****
*
<AzureDiamond> thats neat, I didnt know chatting applications
did that
<Cthon98> yep, no matter how many times you type hunter2, it w
ill show to us as *****
<AzureDiamond> awesome!
<AzureDiamond> wait, how do you know my pw?
/quit
Bye!
```

```
...l on ● SPL in 2025s/w12/week12 via C v13.3.0-gcc took 1m43s
■
```

```
./w12_client localho ~/s/2/w/week12
spl on ● SPL in 2025s/w12/week12 via C v13.3.0-gcc
■ ./w12_client localhost 1398
Enter your username: AzureDiamond
<Cthon98> hey, if you type in your pw, it will show as stars
<Cthon98> ***** see!
hunter2
doesnt look like stars to me
<Cthon98> <AzureDiamond> *****
<Cthon98> thats what I see
... has joined the chat.
oh, really?
<Cthon98> Absolutely
<Cthon98> See, when YOU type hunter2, it shows to us as *****
thats neat, I didnt know chatting applications did that
<Cthon98> yep, no matter how many times you type hunter2, it will show to us as *****
awesome!
wait, how do you know my pw?
... has left the chat.
<Cthon98> er, I just copy pasted YOUR *****'s and it appears to YOU as hunter2 cause its your pw
oh, ok.
```

```
./w12_server 1398 ~/s/2/w/week12
spl on ● SPL in 2025s/w12/week12 via C v13.3.0-gcc
■ ./w12_server 1398
Cthon98 has joined the chat. 1 active user.
AzureDiamond has joined the chat. 2 active users.
got 50 bytes from Cthon98.
got 14 bytes from Cthon98.
got 7 bytes from AzureDiamond.
got 28 bytes from AzureDiamond.
got 22 bytes from Cthon98.
got 16 bytes from Cthon98.
... has joined the chat. 3 active users.
got 11 bytes from AzureDiamond.
got 10 bytes from Cthon98.
got 53 bytes from Cthon98.
got 55 bytes from AzureDiamond.
got 77 bytes from Cthon98.
got 8 bytes from AzureDiamond.
got 28 bytes from AzureDiamond.
... has left the chat. 2 active users.
got 87 bytes from Cthon98.
got 7 bytes from AzureDiamond.
```

Exercise Submission

- **Submit your source code and Makefile in iCampus**
 - Bundle **source code** and **Makefile** with tar command
`$ tar -caf [student_id].tzst week12`
 - We'll grade your submission with **make**
 - » The **make** command should generate **w12_server** and **w12_client** executables.
 - » If compilation fails, your points for this exercise will be zero